

Yeosun Yoo

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EDUCATION

University of California, Davis

Sep 2024 – Jun 2026

M.S. in Computer Science, GPA: 3.870

Davis, CA

Coursework: Machine Learning, Optimization, Math of Data Science, Database Systems, Operating Systems, Program

Verification, Sublinear Algorithms for Big Data

University of California, Davis

Sep 2022 – Jun 2024

B.S. in Computer Science, B.S. in Mathematics GPA: 3.946

Davis, CA

Awards: Dean's Honors List - Fall 22, Spring 23, Winter 24, Spring 24

Coursework: Machine Learning, Algorithms, Operating Systems, Cryptography, Computer Security, Computer Networks,

Numerical Analysis, Mathematical Finance, Stochastic Processes, Real Analysis, Ordinary Differential Equations

PROJECTS & EXPERIENCE

Distributed Metrics Monitoring System | *Rust, Python, Next.js, gRPC*

Spring 2026

- Built a low-overhead telemetry system utilizing Rust data collection agents and a centralized Python backend.
- Designed and implemented a lightweight, structured data serialization protocol via gRPC/Protocol Buffers to minimize network overhead and optimize cross-network payload delivery.
- Developed a clean, real-time visualization dashboard using Next.js to display live performance metrics and historical analytical trends.

Project Lead & President | *LikeLion US Davis Chapter*

Jan 2023 – Jun 2026

- Directed operations for LikeLion, a university software engineering club and tech incubator focused on full-stack development, AI applications, and collaborative product building.
- Led product planning and cross-functional collaboration as Team Leader & Head Engineer in the LLUS Ideathon and Hackathon projects.
- Drove competition outcomes: earned 1st place out of 50+ teams in the 2023 LikeLion Ideathon (sign-language translation app).

Autonomous Vehicle 3D Object Detection Pipeline | *Python, YOLOv8, MiDaS, OpenCV*

Fall 2025

- Architected a vision-only perception pipeline combining 2D object detection (YOLOv8n) and monocular depth estimation (MiDaS) to generate 3D-aware bounding boxes for autonomous driving.
- Engineered a 3D localization framework utilizing pinhole camera geometry to recover physical object dimensions (W, H, D) and map 8-point cuboid spatial orientations from 2D bounding boxes.
- Optimized network performance by fully fine-tuning on the BDD 100K dataset, which doubled Precision to 67.5%, boosted Recall to 41.5%, and raised the overall F1-score to 0.48 compared to a frozen feature extraction baseline.

Performance of Thread Pool Strategies | *C++*

Spring 2025

- Evaluated concurrency strategies (FIFO, LIFO, TBB, OpenMP, OpenBLAS) and native vs. `async/future` models by benchmarking an algorithmically optimized, block-IK matrix multiplication workload.

Handover - Full-stack Developer | *Python, APIs, Vue.js, Github*

Jan 2024

- Developed an AI-powered secondhand marketplace platform utilizing Vue.js frontend and Python/Flask backend.
- Integrated YOLO object detection models to automatically identify and categorize user-uploaded items, improving discoverability.
- Secured 4th place out of 50+ teams at the 2024 Likelion Hackathon by delivering an end-to-end solution under strict time constraints.

TEACHING EXPERIENCE

ECS 189C: Software Correctness with Professor Caleb Stanford, UC Davis

Winter 2026

- Facilitated weekly office hours for 70 students on formal verification, SMT solvers, and automated testing concepts.
- Engineered a Python automated grading system to validate student submissions in Hypothesis, Pytest, Z3 and Dafny, reducing turnaround time by 90%.

ECS 122B: Algorithm Design & Analysis with Dr.Yelena Frid, UC Davis

Spring 2025

- Taught fundamental algorithmic paradigms, complexity bounds, and asymptotic analysis for 150 undergraduate students.
- Created comprehensive grading rubrics and solution keys and provided constructive feedback.
- Moderated the Q&A platform (Piazza), resolving students' confusion and fostering a collaborative learning environment.

TECHNICAL SKILLS

Frontend: JavaScript, React, React Native, TypeScript, Vue.js, HTML/CSS, Tailwind CSS

Backend & Database: Python, MySQL, SQLite, RESTful APIs, Django, Flask, MongoDB, Firebase, Supabase

Data Science & Machine Learning: R, PyTorch, TensorFlow, scikit-learn, keras

Tools & Other Languages: C++, C, C#, GitHub, Cursor, MATLAB, Dart, Go, Rust